Mokhamad Arfan Wicaksono

Data Science, AI/ML, & IoT Researcher and Trainer

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PROFESSIONAL SUMMARY

Researcher and educator specializing in IoT, AI, and Data Science. I lecture, train, and consult across IT domains, with published work in IoT systems and multimedia communications. My focus: applying technology to solve real-world problems.

PROFESSIONAL EXPERIENCE

AI/ML Specialist & Trainer
Jul 2025 - Oct 2025

UrbanSolv, Indonesia

Leading AI and Machine Learning initiatives, providing training and consultation on ML implementation for various projects.

- Machine Learning Trainer for Kementrian PU programs
- Developing AI/ML solutions for urban development projects

Head of Department of Software Engineering Technology
Oct 2024 - Present

Akademi Digital Bandung, Bandung, Indonesia

Leading Software Engineering Technology department, developing curriculum and training programs for digital skills education.

- Leading software engineering education initiatives
- Organizing workshops on Competitive Programming and AI
- Community engagement through BOOST YOUR BUSINESS programs

Head of Knowledge and Expert
Nov 2023 - Oct 2025

Enclave Kolaborasi Global (EKG), PT, Indonesia

Managing knowledge initiatives and providing expert consultation on data analytics and AI projects.

- Developed AI-based Proposal Checker Platform
- Conducted Data Analytics workshops for government and corporate clients

Academic Assistant
Aug 2023 - Dec 2024

KKTI STEI ITB, Bandung, Indonesia

Assisting in academic activities and research support at School of Electrical Engineering and Informatics.

- Supporting academic programs and student activities
- Contributing to research projects

Researcher
Apr 2019 - Dec 2024

DSP-MC Laboratory ITB, Bandung, Indonesia

Conducting research on Wireless Sensor Networks, Multimedia IoT, and Machine Learning applications.

- Published multiple papers on IoT and Multimedia communications
- Led research on CoAP communication methods for IoT
- Developed ML-based video delivery optimization systems

Research and Development Intern
May 2017 - Aug 2017

PT Indo Cis Cisco Network Security Consultant, Indonesia

Internship focused on network security research and development projects.

- Conducted research on network security solutions
- Contributed to R&D initiatives

EDUCATION

Master of Science in Informatics
Aug 2019 - Dec 2021

Bandung Institute of Technology, Bandung, Indonesia
GPA: 3.51/4

- Research focus on Wireless Sensor Networks and IoT
- Published multiple research papers on Multimedia IoT
- PPMI Research Grant recipient

Bachelor of Science in Information System and Technology
Aug 2014 - Jul 2019

Bandung Institute of Technology, Bandung, Indonesia
GPA: 3.72/4

- Final Project: Development of Cat Care System Based on Internet of Things
- Silver Prize at d-Camp 2018 Engineering Solution Contest
- Active in organizational and committee activities

High School Diploma in Natural Science Major
Jul 2011 - Jun 2014

SMA Negeri 15 Bandung, Bandung, Indonesia
GPA: 50.7/60

Junior High School Diploma in General Education
Jul 2008 - Jun 2011

SMP Negeri 2 Bandung, Bandung, Indonesia
GPA: 35.95/40

CERTIFICATIONS

IELTS Band 7

British Council | Feb 2025 - Feb 2027

Credential ID: 24ID516923WICM195A

Sertifikat EF SET Bahasa Inggris 81/100 (C2 Mahir)

EF SET | Jan 2025

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Deep Learning & CNN, Computer Vision, Data Science & Analytics, Agentic Coding

- **Machine Learning**
  5+ years | Scikit-learn, TensorFlow, PyTorch, XGBoost
  – Built ML-based frame resolution adjustment for IoT video optimization
  – Developed predictive models for multimedia delivery systems
  – Published research in ACM Transactions on ML applications
- **Deep Learning & CNN**
  4+ years | TensorFlow, Keras, PyTorch, Inception-V3
  – Developed CNN models for music genre classification
  – Built computer vision systems for plant disease detection
  – Created garbage classification system using Inception-V3 and SVM
- **Computer Vision**
  4+ years | OpenCV, CNN, RFID, Python
  – Developing AIoT attendance system with CV and RFID
  – Built image processing for hydroponic plant monitoring
  – Implemented real-time object detection systems
- **Data Science & Analytics**
  6+ years | Python, Pandas, NumPy, Matplotlib, SQL
  – Conducted data analytics workshops for government agencies
  – Analyzed QoE metrics for multimedia IoT systems
  – Built data pipelines for IoT sensor networks
- **Agentic Coding**
  2+ years | GitHub Copilot, Claude Code, GLM Code, Claude Haiku, Claude Sonnet, Claude Opus, GPT Codex
  – Leverage AI coding agents for rapid prototyping and development

– Utilize multiple LLM models for code generation and review
– Implement agentic workflows for automated testing and debugging
– Integrate AI pair programming into daily development workflow

Internet of Things: IoT System Architecture, RFID Technology, Wireless Sensor Networks, Embedded Systems

- **IoT System Architecture**
  7+ years | CoAP, MQTT, WSN, Edge Computing, Cloud IoT
  – Designed software-defined wireless multimedia sensor networks
  – Built end-to-end IoT solutions from sensors to cloud
  – Published 10+ research papers on IoT systems optimization
- **RFID Technology**
  3+ years | RFID Readers, NFC, IoT Integration, Computer Vision
  – Developing campus attendance system with RFID and CV
  – Integrated RFID with QoE and accountability measures
- **Wireless Sensor Networks**
  6+ years | CoAP, 6LoWPAN, SDN, WSN Protocols
  – Optimized multimedia delivery on WSN for IoT
  – Developed SDN-based WSN management systems
  – Published research on WSN QoS improvement
- **Embedded Systems**
  6+ years | Arduino, ESP32, Raspberry Pi, Sensors, Actuators
  – Built cat care system with automated feeding and monitoring
  – Developed hydroponic monitoring with environmental sensors
  – Created smart agriculture IoT devices

Programming & Development: Python, Web Development, Database Management

- **Python**
  7+ years | Django, FastAPI, NumPy, Pandas, Matplotlib, TensorFlow
  – Built ML pipelines and data processing systems
  – Developed APIs for IoT data collection and analysis
  – Created AI-based proposal checker platform
- **Web Development**
  5+ years | JavaScript, React, Node.js, REST APIs
  – Built responsive web applications for IoT monitoring
  – Developed dashboards for real-time sensor data visualization
  – Created educational platforms and competition systems
- **Database Management**
  6+ years | MySQL, PostgreSQL, MongoDB, Redis
  – Designed databases for IoT time-series data storage
  – Optimized queries for real-time analytics
  – Implemented data pipelines for sensor networks

Research & Education: Academic Research, Teaching & Training, Project Management

- **Academic Research**
  7+ years | LaTeX, Research Methodology, Statistical Analysis
  – Published 15+ papers in international journals and conferences
  – Led 6+ funded research projects (national and institutional)
  – Expertise in IoT, ML, and WSN research domains
- **Teaching & Training**
  5+ years | Curriculum Development, Workshop Design, Mentoring
  – Head of Software Engineering Technology Department
  – Trained 100+ professionals in ML, IoT, and Data Science
  – Delivered workshops for Samsung, government agencies, and universities
- **Project Management**
  5+ years | Agile, Scrum, Research Planning, Team Leadership

– Led multi-institutional research collaborations
– Managed knowledge initiatives and expert consultations
– Organized technical competitions and educational programs

RESEARCH PROJECTS

Sistem Absensi Kampus Terintegrasi Berbasis AIoT dengan Pendekatan QoE dan Akuntabilitas Menggunakan RFID dan Computer Vision

Penelitian Dosen Pemula Kemendikbudristek | 2025 | Ongoing

Integrasi Agentic AI dalam Pembelajaran Adaptif untuk Meningkatkan Kompetensi Bisnis Digital Mahasiswa

Penelitian Dosen Pemula Kemendikbudristek | 2025 | Ongoing

Peningkatan Kinerja Pengiriman Multimedia pada Internet of Things Berbasis Jaringan Sensor Nirkabel

Penelitian Kompetitif Nasional | 2023 | Completed

Pengembangan Sistem Pengiriman Data Berbasis Machine Learning untuk Peningkatan QoE pada Multimedia Internet of Things

Riset dan Inovasi untuk Indonesia Maju | 2023 | Completed

Sistem Pengelolaan WSN pada Multimedia IoT untuk Peningkatan Quality of Service

PPMI | 2021 | Completed

Software-Defined Network pada Jaringan Sensor Nirkabel untuk Sistem Multimedia-Internet of Things (M-IoT)

Riset ITB | 2021 | Completed

Pengembangan Platform Perawatan Tanaman Hidroponik dengan Analisis Citra Digital Berbasis Algoritma CNN dan IoT

Penelitian Terapan Unggulan Perguruan Tinggi | 2020 | Completed

PUBLICATIONS

"The New Educational Frontier: Agentic AI's Evolutionary Journey Through The Lens of SPAR-4-SLR"

VOCATECH: Vocational Education and Technology Journal, Dec 2025

"Performance Evaluation of CoAP Communication Method Extensions in Internet of Video Things"

Journal of Advances in Information Technology, Jul 2025

"Music Genre Classification Using Convolutional Neural Network Method"

2024 International Conference on Information Technology Systems and Innovation (ICITSI), Mar 2025

"Arsitektur Software-Defined Wireless Multimedia Sensor Network (SDWSN) Berbasis CoAP untuk Optimasi Komunikasi dan Manajemen Jaringan"

DISTech: Digital Innovation and Sustainable Technology Journal, Jan 2025

"IoT Video Delivery Optimization Through Machine Learning-Based Frame Resolution Adjustment"

ACM Transactions on Multimedia Computing, Communications, and Applications, May 2024

"Combining inception-V3 and support vector machine for garbage classification"

Jurnal Infotel, Nov 2023

"Recyclable Waste Classification using SquezeeNet and XGBoost"

(IJACSA) International Journal of Advanced Computer Science and Applications, Jan 2023

"CoAP Evaluation for Node-Edge Communication on Wireless Multimedia Sensor Network"

2021 International Conference on Sustainable Engineering and Creative Computing, Dec 2021

"Development of Cat Care System Based on Internet of Things"

2019 International Conference on Electrical Engineering and Informatics, ICEEI, Feb 2020

INTELLECTUAL PROPERTY

HydroponicSense - Computer Program Copyright, Indonesia Ministry of Law (Dec 2020)

Cat Care - Computer Program Copyright, Indonesia Ministry of Law (Mar 2019)

PROJECTS

RFID-based Student Attendance Platform
IoT-based campus attendance system using RFID and Computer Vision with QoE and accountability approach.
Expertise: IoT, RFID, Computer Vision, Python, Machine Learning

AI-based Proposal Checker Platform
AI platform for checking and evaluating proposals.
Expertise: AI, NLP, Python, Machine Learning

Cat Care System Based on Internet of Things
Internet of Things-based cat care system for automated pet monitoring and feeding.
Expertise: IoT, Arduino, Sensors, Mobile App, Cloud

Enterprise Architect for Radio-B
Designing an Enterprise Architect for Radio-B.
Expertise: Enterprise Architecture, Business Process, System Design

Integrated Health Information System
Designing integrated health information system for healthcare facilities.
Expertise: System Design, Healthcare IT, Database Design, Multimedia

Web Information System for Siete Cafe
Designing web-based information system for Siete Cafe.
Expertise: Web Design, Requirement Analysis, System Design, Database

Online Delivery System for Mulya Cirebon Chinese Food
Designing online delivery system for Chinese food restaurant.
Expertise: System Design, Service Design, Database, Web

Web-Based Hotel Reservation and Service System
Develop web-based hotel reservation and service management system.
Expertise: Web Development, PHP, MySQL, Software Engineering

Student Monitoring System for Salman Al-Farisi Primary School
Designing student monitoring and management system for primary school.
Expertise: System Design, Requirement Analysis, Database Design

Online Booking System for Futsal Field Reservation
Planning online booking system project for futsal field reservation.
Expertise: Project Management, System Planning, Scheduling

Harvest Moon Game Replica with Java
Develop replica of Harvest Moon game using Java programming.
Expertise: Java, OOP, Game Development

Flight Database Management System
Designing flight database management system for airline operations.
Expertise: Database Design, SQL, Data Modeling, ERD

MHoney Website
Web application for MHoney (mhoney.id) built with React and Tailwind CSS.
Expertise: React, Tailwind CSS, SEO, Google Analytics

Invoice Generated
AI-powered invoice generation platform built with React, Hono, and Cloudflare.
Expertise: React, Hono, Cloudflare Pages, Cloudflare Workers, AI Integration

AWARDS & ACHIEVEMENTS

2nd Winner of Business IT-Case Competition
Techphoria Event | Universitas Sriwijaya | National | Oct 2017

Silver Prize - Engineering Solution Contest
d-Camp 2018 | Innovation Hub Center for Engineering Education (IHCEE) Chonbuk National University | International | Aug 2018

Participant Prize - i-CAPS 2018
International Students Joint Capstone Design Project Competition | Innovation Hub Center for Engineering Education (IHCEE) Chonbuk National University | International | Aug 2018

ORGANIZATIONS

Head of Human Resource Management
Lingkung Seni Sunda (LSS) ITB | Sep 2016 - Sep 2017

Vice Head of Intra-Campus Division
Himpunan Mahasiswa Informatika (HMIF) ITB | Feb 2015 - Jan 2016

COMMITTEE EXPERIENCE

Head of Logistics Division - Arkavidia Informatics and IT Festival
HMIF ITB | 2016

Chairman - HMIF July Graduation 2016
HMIF ITB | 2016

General Secretary - New LSS ITB Member Recruitment 2015
LSS ITB | 2015

TRAINING & SPEAKING ENGAGEMENTS

IoT Instructor — Oct 2025
Samsung Innovation Campus x dibimbing.id

IoT Mentor — Sep 2025
Samsung Innovation Campus x dibimbing.id

Speaker at Workshop AI & Data Science — Aug 2025
Pengmas Akademi Digital Bandung

Speaker at AI and IoT Webinar — Aug 2025
Samsung Innovation Campus x dibimbing.id

Machine Learning Trainer — Jul 2025
Kementrian PU x UrbanSolv

IoT Workshop Trainer — Jun 2025
Kementrian Ketenagakerjaan x dibimbing.id

Private IoT and Data Science Teacher — May 2025
UNSIA Student

Competitive Programming Workshop Trainer — Mar 2025
Akademi Digital Bandung

Basic Programming Workshop Trainer — Mar 2025
Personal Charity Project

Speaker at BOOST YOUR BUSINESS: AI UNTUK UMKM NAIK KELAS — Dec 2024
Pengmas Akademi Digital Bandung

Conference Moderator — Sep 2024
International Conference on ICT for Smart Society by ITB

Data Analytics Workshop Trainer — Jun 2024
PT Cerebrum Edukasiesia Nusantara

Data Analytics Workshop Trainer — Mar 2024
Jasa Raharja BOD Candidate by Linxbrain

Data Analytics Workshop Trainer — Dec 2023
Diskominfo Employee by Enclave Kolaborasi Global

Data Analytics Workshop Trainer — Sep 2023
FEUNM

Data Analytics Workshop Trainer — Aug 2023
Diskominfo Employee

Webinar Speaker in AI Topic — Jul 2023
UNIBI

IoT Trainer — Feb 2023
SMKN 2 Sukoharjo

Back Office Trainer — Jan 2023
Daarut Tauhid Boarding School for Female

Data Science Trainer — Aug 2022
HMP PL ITB Komisariat 2022/2023

Excel Trainer Nov 2020
Daarut Tauhid Boarding School Civitas Upgrading

IoT Facilitator Sep 2019
Digital Talent Program by Kominfo at ITB

IoT Facilitator Jul 2019
Digital Talent Program by Kominfo at ITB

FGD Speaker: Fundamental AI and IoT Oct 2025
Politeknik Negeri Cilacap